Data Structures Notes - Units 1 to 3

College Buddy - free study resource sample file

Unit 1: Introduction and Arrays

- Time complexity: Big-O, Big-Omega and Big-Theta notations.
- Array operations: insertion, deletion and traversal with examples.

Unit 2: Stacks and Queues

- Stack using arrays and linked lists; push, pop and peek operations.
- Circular queue, deque and priority queue with applications.
- Infix to postfix conversion using a stack (worked example).

Unit 3: Linked Lists

- Singly, doubly and circular linked lists with diagrams.
- Reversing a linked list (iterative and recursive methods).
- Important viva questions: why linked lists need no contiguous memory?